Whole-Home Low-Voltage + Load-Control Installation Standard

KNX + DALI-2 (Residential)

Project: Residential Retrofit/New Build (Contra Costa County, CA)
Document Purpose: Single standard for boxes, cabling, mixing rules, and load-control patterns
Revision: v2.7 (Environmental Monitoring & PM2.5)
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0. Desired Outcomes and Design Rationale

This document intentionally avoids an 'approved/not approved' posture. If an alternate box/cable/device is preferred by the electrician, use the vetting checklist in Section 9 (Listing, ratings, fit, and DALI-2 Verified for any device on the DALI bus) and record the final choice in the project BOM/as-built set.

Deviation handling:

- **Owner:** will provide device registration, addressing, and programming/commissioning for the automation system (DALI short addresses, group/scenes, KNX programming), and will maintain the as-built documentation. **All logic and software configuration is the responsibility of the Owner.**
- **Electrician:** propose practical, code-compliant implementation details and equivalent listed products where needed; coordinate routing, box fill, conductor derating, and all line-voltage terminations. **Does NOT include system programming.**

Roles and collaboration:

- Standardize repeated patterns across the house (same box families, consistent labeling, repeatable LCP layout) to reduce install risk and simplify maintenance.
- Prefer DALI-2 daisy chaining/branching where it reduces copper runs and improves flexibility, while prohibiting DALI loops (no rings/meshes).
- Prefer open standards and open procurement. KNX and DALI-2 are used because they are interoperable, multi-vendor ecosystems. Lutron is not used because it is a walled garden.
- Dimming everywhere for lighting loads. Decorative fixtures with replaceable bulbs are acceptable; lamp selection can be adjusted by the Owner to achieve the desired dimming quality.
- Full home automation for all electrical loads (lighting, fans, exhaust, heaters, and other fixed equipment), with each load individually addressable and controllable.

Desired outcomes: This document communicates the Owner's project goals and the preferred implementation patterns for a whole-home KNX + DALI-2 system. It is intended as an input to the electrician for (a) coordinated rough-in decisions and (b) a cost estimate. The electrician remains the expert of record for code compliance, installation methods, and final equipment selection.

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1. Scope and Non-Negotiables

1.1 Scope

Applies to all KNX and DALI-2 load-control work in this home, including any junctions or ceiling/wall boxes that contain control modules or mixed-voltage splices.

1.2 Non-negotiables (Hard NO list)

- Standardize on the box family, ceiling box standard, and DUO cable method. If an alternate is proposed, validate it against Section 9 and record the decision in the project BOM.
- **No plastic boxes** at mixed-voltage load-control locations.
- **No non-fan-rated boxes** at ceiling outlets that could reasonably receive a ceiling fan now or in the future.
- **No field-created DUO bundles** (taping separate cables together). Use only listed power+control assemblies where required.
- **No DALI bus loops** (ring/mesh). Branching is allowed; closed loops are not.

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2. Code Basis and Inspection Philosophy (California)

Work must pass inspection under the California Electrical Code (CEC) as adopted by the local AHJ. The licensed electrician remains responsible for final box-fill calculations, conductor derating, and enforcing the manufacturer instructions and listings for every installed component.

Key inspection themes this standard is designed to satisfy:

- Listed and labeled equipment used per instructions (NEC/CEC 110.3(B)).
- All junctions remain accessible (NEC/CEC 314.29).
- Appropriate separation or approved mixing method for Class 1 and Class 2/Class 3 conductors (NEC/CEC 725).
- Ceiling fan support boxes are listed and marked for fan support where applicable (NEC/CEC 314.27).

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3. Box and Enclosure Standards

3.1 Box Standard (House Default)

To maximize consistency, serviceability, and future flexibility, standardize on roomy UL/cULus Listed steel boxes and a small set of accessories across the entire house.

- **Default box:** Use a 4-inch square, ~2-1/8-inch deep steel box (or closest readily-available equivalent) for most junctions/rough-in locations, including in-wall (2×4 / 3.5-inch studs) and attic work.
- **Downsize by opening, not by box:** Use mud rings/plaster rings to adapt the same box to 1-gang, 2-gang, or fixture openings instead of changing box families.
- **When in doubt, add volume:** Use extension rings (or a larger box) rather than overfilling a small box.
- **Safety note:** Any ceiling outlet that may support a fan must use a listed fan-rated box appropriate for the installation.

3.3 Mixed-voltage boxes (line voltage + DALI/KNX in the same box)

This project allows and often requires mixed boxes (example: DUO cable in, DALI-2 relay module in-box, line-voltage load out).

- **Preferred default:** all DALI/KNX conductors present in any box that also contains line voltage are 600V-rated (or remain inside a listed 600V-rated assembly such as MC-PCS Duo).
- **Alternate:** use a listed barrier/partition system (example: RACO 885 partition cover) to separate compartments. If a listed barrier is not available for that configuration, use a separate box.

- Neat dressing, bonding/grounding, and no exposed SELV-only insulation are required in mixed boxes.

3.4 UL Recognized modules inside listed boxes

UL Recognized (UR) control modules are allowed only when installed inside a UL/cULus listed box per the module manufacturer instructions and Conditions of Acceptability (COA). Provide documentation to the AHJ if requested.

3.5 Load Control Panels (LCPs)

Purpose: Provide a clean, serviceable, repeatable "control node" near groups of loads (especially high-current or many LED strip loads). **LCP requirements (Office / recessed LCP-2):**

- Max cabinet depth: 4.0 in total installed depth (3.5 in stud + 0.5 in drywall).
- Hardware mounting: All devices installed inside LCP-2 MUST be DIN-rail mountable, and shall fit within the 4.0 in depth with wiring and bend radius.
- Safety listing: Enclosure must be UL 50/50E Listed (Type 1) or equivalent; maintain line-voltage vs control separation per NEC using listed barriers/ducting as needed.
- Cable strategy: Bring HV feed(s) in, then distribute as needed. Bring DUO (power+control) or separate conductors out to loads; label every departure.

Office LCP-2 physical constraints (between studs): 14.5 in clear stud bay width; 39 in max height below fire-block. Recommended standard enclosure sizes:

- nVent HOFFMAN ASE24X12X4NK (24×12×4 in, Type 1, UL File E27525)
- nVent HOFFMAN ASE12X12X4 (12×12×4 in, Type 1; use as auxiliary expansion below 39 in)

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4. Cable, Splices, and Labeling

4.1 DUO / combined power + control cable

Where power and DALI/KNX control share the same pathway or box, cable must be a listed power+control assembly.

- **Southwire MC-PCS Duo (Type MC)** for armored runs.
- **Southwire NM-B-PCS Duo (Romex)** where NM cable is permitted.
- All conductors are 600V-rated by listing.

4.2 Approved splice/connect methods

Use listed connectors rated for the conductor sizes and voltage present. In mixed boxes, treat all splices as 600V wiring method.

Use case	Approved connector family	Typical conductor range	Notes
DALI/KNX pair splices	WAGO 221 series (lever nuts)	24-12 AWG copper	600V rated; keep DA pair together
Branch-circuit power splices	WAGO 221 series or wirenuts	12/14 AWG copper	Match wire count and gauge
Ground/bond splices	Listed grounding pigtail + WAGO	Per EGC size	Bond box; use green screws

4.3 Labeling (Required)

Label every box and every DUO cable run with printed heat-shrink or durable wrap labels.

- Branch circuit ID (panel/breaker, circuit number).
- Load served (room + device).
- DALI universe/line ID and direction (Upstream/Branch A/Branch B).

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5. DALI-2 Bus Architecture and Topology

5.1 Design limits

- 64 addressed devices per bus.
- Bus power supply current up to 250 mA.
- Practical line length around 300 m with 16 AWG conductors.

5.2 Allowed vs prohibited topologies

- **Allowed:** line, tree, and star.
- **Prohibited:** any topology that creates a closed loop/ring/mesh.

5.3 Standard "DALI Tree Junction" node (T-junction)

Purpose: create a clean way to split a DALI trunk into two downstream branches.

- Use a 4-11/16 inch square steel box (RACO 254/257).
- Bring in 1 upstream DUO cable and 2 downstream DUO cables.
- Splice power conductors per normal practice.
- Splice DALI conductors DA/DA across all three cables using WAGO 221 (600V).
- **Wiring diagram (conceptual):** `[Upstream trunk]----(Node)----[Branch A]` with `[Branch B]` connected at the node.

5.4 Standard "DALI Tree Junction" node (4-way / X-junction)

Use when you need 3 downstream branches. Same box and method as T-junction. Avoid large stars; keep branching deliberate.

5.5 Loop prevention and field verification

- **As-built tree diagram:** every DALI run has exactly one upstream source.
- **Continuity sanity check:** with bus power disconnected, verify no cross-connection between branches.
- **Label audit:** every junction node has labeled branches.

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6. Approved Control Hardware (Vendor-Neutral)

6.0 Universal Dimming Requirement

- **DALI-2 verification (required):** Any device connected to the DALI bus MUST be listed in the DiiA DALI-2 Product Database.
- **Safety listing:** Device must be UL/cULus Listed or UL Recognized (with COA).

Table 6-1: Benchmark Lighting Control Devices

Category	Benchmark Device	DALI-2 (DiiA)	UL/ETL Status	Notes
Phase dimming	Sunricher SR-2303SAC-HP	ID 5831	File E477171	Micro module; watch min-load
CC Driver	eldoLED SOLOdrive 360/S	ID 371	UR Component	0.1% class; flicker-free
LED Strips	Lunatone DT8 Dimmer + Mean Well PSU	ID 10471	Verified Listed	Centralized in LCP
0-10V bridge	Sunricher SR-2401-10V	ID 3065	File E477171	Signal-only; keep relay separate

Table 6-2: Power & Switching Loads

Category	Benchmark Device	UL Evidence	Relay Rating	Bus Load
Relay Puck	SR-2701S-DT7	File E477171	4.3A Resistive	18 mA
4-ch DIN Relay	SR-2701DIN-DT7	File E477171	12A per ch	2 mA

Switching Module Comparison Table:

Category	Example model	Function	Listing	Notes
DALI-2 -> 0-10V + relay	NICOR IMS3D0-10VRLY	DT5 + DT7	cULus Listed	Default for 0-10V
DALI-2 -> 0-10V + relay	LOYTEC LDALI-RM6	DT5 + DT7	cULus Listed	1/2-inch nipple box mount
DALI-2 relay-only	Wattstopper WA-DALI-SWITCH-SO	DT7	UL Listed	3-channel relay
High-current relay	Functional Devices RIB2401B	External	UL Listed	Use for motors; 1/2-inch nipple
Contactor	Eaton C25 series	Contactor	UL Recognized	Use in LCP for 40-50A class

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7. Standard Installation Patterns

7.1 Pattern A: 0-10V dimmable driver/fixture + hard on/off

Use DALI-2 to 0-10V converter with integrated relay local to the load.

7.2 Pattern B: Relay-only on/off

Use DALI-2 relay module. For motors, drive a motor-rated relay (RIB) or contactor coil.

7.3 Pattern C: High-current loads (heaters, large motors)

Place the switching device in the nearest LCP (DIN rail). Use DALI-2 relay output to control a contactor/relay coil. **LCP depth limit is 4.0 inches.**

7.5 Ceiling fans and fixtures

Ceiling outlet uses the fan-rated box standard. Canopy volume must be confirmed for control modules.

7.6 Pattern E: LED strip lighting (DT8 tunable white)

Centralized drivers in LCP. Minimum zoning: one zone per room. Smooth dimming to low levels required.

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8. HVAC Coordination

- HVAC Class 2 control wiring must not share boxes with line voltage unless 600V-rated.
- Switched line power (exhaust fans, heaters) must use rated relay/contactor.
- All HVAC-related junctions must remain accessible (do not bury in insulation).

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9. Vendor and Part Vetting Process (Required)

1. **Listing status:** UL/cULus Listed preferred.
2. **Ratings:** Confirm resistive and inductive/motor ratings at 120/240/277V.
3. **Mechanical fit:** Confirm fit in approved box with bend radius.
4. **DALI-2 compatibility:** REQUIRED (DiiA entry).
5. **Open procurement:** Confirm open-market availability.
6. **Documentation packet:** Store datasheet + instructions + listing evidence.

Appendix A. Field Checklists

A1. Electrician checklist (rough-in)

- 4-11/16 inch steel boxes + listed rings/covers used.
- Fan-rated boxes at all ceiling outlets.
- 600V-rated control conductors in mixed boxes.
- DALI junction nodes labeled and accessible.

A2. Owner/commissioning checklist

- Record DALI addresses and branch node IDs.
- Confirm bus current/address limits.
- Photo capture of every junction node before closing.

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10. Model Selection (Standard Units)

This section defines specific hardware models selected as the "Project Standard." To maintain architectural consistency and minimize visual clutter, these models must be used for all repeated load types unless a specific deviation is recorded in the project BOM. This specifies the models only, not the quantities or locations.

Programming Note: All device addressing, bus commissioning, and logical grouping (KNX/DALI) will be performed by the Owner.

10.1 Recessed "Canned" Lighting (Standard Downlight)

The project standard for all recessed architectural downlighting is the **DMF X-Series**. This system is selected for its modularity and high-quality DALI-2 integration.

- **Visual Goal:** Minimize the visual aspect of the fixture on the ceiling.
- **Specified Trim:** Use the **White insert / reflector** (`X2TSDSWHFL`) to provide a clean, integrated look that minimizes the visual presence of the light aperture.
- **Specified Components:**
  - **Housing:** `X2NCS` (New Construction Square Housing).
  - **Mud-in Plate:** `X2KSMUD` (Square Flangeless Mud-In Kit).
  - **Trim:** `X2TSDSWHFL` (Square Flangeless Trim with White Insert).
  - **Light Module:** `XMD` (DALI-2 LED Module, 1250lm, 3000K, Wide Flood).

10.2 Ceiling Fans

The project standard for ceiling fans is the **Big Ass Fans Haiku**.

- **Control Interface:** Every fan unit must be equipped with the **0-10V control interface** (Part Number: **010447** for current Haiku models) for variable speed and light integration.
- **Wiring Requirements:** Two separate 0-10V analog channels are required per fan:
  - **Channel 1 (Fan Speed):** 10V = 100% speed, 0V = Off.
  - **Channel 2 (Light Intensity):** 10V = 100% brightness, 0V = Off.

- **Bridge Integration:** Both signals are bridged to the DALI-2 bus using listed DALI-to-0-10V bridges (benchmark: Lunatone/Sunricher). Note that installing the 010447 module disables the factory remote and mobile app control in favor of the automation system.
- **Specified Units (Two Versions):**
  - **Haiku (No LED Light):** Standard fan assembly. Only the Fan Speed channel is utilized.
  - **Haiku (with LED Light):** Fan assembly with integrated LED. Both Speed and Light channels are utilized.

10.3 Bathroom and Laundry Sensors (Environmental Multisensors)

The project standard for indoor environmental sensing and presence detection in bathrooms, toilets, and the laundry room is the **Steinel True Presence Multisensor KNX**.

- **Model Specification:** Steinel True Presence Multisensor KNX (Flush Version).
  - **Article Numbers:** #056353 (White) or #086060 (Black).
  - **Sensors:** True Presence (radar-based), VOC (odors/volatiles), CO_2, Humidity, Temperature, Air Pressure, and Brightness.
  - **Power:** Low-voltage KNX bus-powered only.

- **Visual & Mechanical Integration:**
  - **Mounting:** Flush ceiling install using standard US 4-inch round remodel boxes (or equivalent with depth ≥33 mm).
  - **Placement Logic:** Position sensors centrally or offset directly near/under the **linear architectural exhaust diffusers**. This ensures optimal capture of rising odors and humidity while assisting the airflow "updraft" sensing.

- **Specific Locations:**
  - **Master Bathroom Toilet Room:** Above/near the toilet, adjacent to the linear diffuser.
  - **Master Bathroom Main Area:** Near the shower/tub linear exhaust (radar ensures lights/fans stay on during stationary shower use).
  - **Dedicated Guest Toilets:** Above the toilet / near the linear diffuser.

    - **Laundry Room:** Positioned near the cat litter box and primary exhaust vent to handle ammonia and humidity spikes.
- **Control Logic (Ventilation Integration):**
    - **Presence Detection:** Triggers the exhaust fan to "Low/Quiet" speed.
    - **Environmental Thresholds:** High VOC, CO_2, or Humidity levels trigger the fan to "High/Boost" speed until levels return to the setpoint baseline.
- **System Architecture:**
    - Assign physical addresses hierarchically across the Load Control Panels (LCPs).
    - Share group addresses for coordinated fan control across the KNX/DALI bridge.
    - Use Line Couplers or IP Routers for reliable communication between panels.

10.4 Whole-House Fan (Ducted, Attic-Mounted, High-Efficiency)

The project standard for whole-house cooling is the **QuietCool Trident Pro 7.0X**.

- **Model Specification:** QuietCool Trident Pro 7.0X (EC Motor, variable speed via 0-10V).
- **Quantity: 2** (One per arm of U-shaped layout for even airflow; total 14,000 CFM over-spec for 4,200 sq ft house).
- **Pricing:** Unit Price ≈ $1,250 each.
- **DALI-2 Integration Components (Per Fan):**
    - **Relay:** Sunricher SR-2701S-DT7 (DALI-2 Relay, 300W/240V AC rated): $110 each.
    - **Converter:** Sunricher SR-2401-10V (DALI-2 to 0-10V signal): $65 each.
- **Total Integration Qty:** 2 relays + 2 converters.
- **Operation Notes:**
    - **Acoustics:** 45-55 dBA on low-high; typically run at 50-70% for maximum quiet/efficiency.
    - **Mounting:** Hung mounting with included vibration isolation kit.
    - **ETS Logic:** Program for low-speed baseline (4-6V) and high-boost (10V) on thresholds/interlocks.

10.5 Attic Exhaust Fan (Gable-Mounted, Variable Speed)

Standardized gable exhaust to assist whole-house airflow and manage attic heat.

- **Model Specification:** QuietCool AFG PRO-3.0 (EC Motor, variable speed via 0-10V).
- **Quantity: 4** (Two per gable end for balanced exhaust; total 12,000 CFM over-spec).
- **Pricing:** Unit Price ≈ $450 each.
- **DALI-2 Integration Components (Per Fan):**
    - **Relay:** Sunricher SR-2701S-DT7: $110 each.
    - **Converter:** Sunricher SR-2401-10V: $65 each.
- **Total Integration Qty:** 4 relays + 4 converters.
- **Operation Notes:**
    - **Acoustics:** Ultra-quiet 35-45 dBA.
    - **Interlock:** Program via ETS to assist whole-house fans (e.g., 50% speed assist) to prevent backpressure and backdrafting.

10.6 Motorized Skylight System (Standardized Integration)

The project utilizes a centralized KNX-based approach to control motorized skylights across the home, ensuring architectural consistency and integration with the ventilation/HVAC logic.

- **Units (8 Total):**
    - **Hallway:** 1 × Velux VSE M08 (~2'×4')

  - **Dining Room:** 6 × Velux VSE C01 (~2'×2')
  - **Laundry:** 1 × Velux VSE C01 (~2'×2')
- **Control Groups (3 Logical Zones):**
  - **Group 1 (Hallway):** 1 skylight.
  - **Group 2 (Dining Room):** 6 skylights (daisy-chained).
  - **Group 3 (Laundry):** 1 skylight.
- **Control & Feedback Hardware (LCP-Resident):**
  - **KNX Actuator:** MDT JAL-0810M.02 (8-channel, 24V DC). Controls 3 groups via polarity reversal (Open/Close/Stop).
  - **Power Supply:** Mean Well MDR-100-24 (100W, DIN-rail). Provides 24V DC bus for skylight motors.
  - **Binary Input:** MDT BE-08000.02 (8-fold). Receives 3 reed switch feedback signals for state confirmation.
- **Feedback Mechanism:**
  - **Sensors:** Surface-mount reed / magnetic contact switches (8 total, wired in parallel per group).
  - **Single Point Feedback:** Each group returns a single feedback pair to the binary input (Open vs Closed).
- **Cabling Strategy:**
  - 3 × Southwire MC-PCS Duo / NM-B-PCS Duo home-run from LCP (one per group).
  - Power and control signals share the primary conductors; the LV pair carries the parallel reed switch feedback loop.
- **Operation & Safety Logic:**
  - **Position Tracking:** Determined by calibrated travel time with absolute confirmation from reed switches at end-of-travel.
  - **Fault Detection:** If a Close command is sent but the reed switch still reports Open after a timeout, the ETS logic flags an error (alert, retry, or safety HVAC interlock).
  - **Interlock Logic:** ETS groups facilitate priority control for weather safety (rain sensors), fan interlocks, and HVAC energy conservation.

10.7 Weather Station (KNX Secure)

The project utilizes the **Warema KNX secure Weather station pro REG** (Article No. #2064965) for real-time environmental data and automation triggers.

- **Model Specification:** Warema Weather Station Pro REG (DIN-rail module + outdoor sensor head).
  - **Sensors:** Ultrasonic wind speed/direction, optical rain/snow, 360° brightness + twilight, global radiation, temperature, and GPS (sun position/time).
  - **Compliance:** KNX Secure certified; no moving parts (solid-state).
- **Physical Integration:**
  - **Outdoor Sensor:** Compact dome mounted to the **North-facing wall** on the left side of the house.
  - **Mounting Height: 9–10 feet** above ground level (serviceable from a standard 6-foot ladder).
  - **Indoor Module:** DIN-rail mounted in **LCP-2 (Office)**.
  - **Power:** 24V DC (~13mA max). Supplied via KNX bus or dedicated DR-15-24 PSU.
- **Cabling:** UV-resistant 4-conductor cable (AWG 18–20) home-run to LCP-2 (~10–65 ft typical).
- **Placement Logic:** Avoid deep eaves to ensure clear "sky view" for precipitation and brightness sensors.

10.8 Outdoor Air Quality Sensor (Wildfire PM2.5 Focus)

Real-time PM2.5 monitoring for smoke detection to protect indoor air quality during wildfire events.

- **Model Specification: Temco Outdoor Air Lab PM2.5 Sensor (OAL-PM2.5)**.
  - **Interface:** RS485 Modbus RTU.
  - **Power:** 15–24 V DC/AC (~2 W).
- **Integration Hardware (LCP-2):**
  - **KNX/Modbus Gateway:** MDT SCN-MB.01.
  - **Power Supply:** Mean Well DR-15-24 (dedicated 24V rail).
- **Physical Integration & Cabling:**
  - **Location:** Left side of the house, North-facing wall-mounted at **5–7 feet** above ground level.
  - **Cabling:** One Cat6 cable home-run to LCP-2. Pair 1 for Modbus Data (A/B), Pair 2 for 24V Power.
  - **Placement Logic:** Protected from direct rain/sun but with adequate airflow; away from vents or exhaust outlets.
- **Approx. Pricing:** Sensor (~~$250) + MDT Gateway (~~$280).

10.9 Data Flow & Automation Logic (Environmental)

All environmental data terminates in LCP-2 (Office) and is exposed to the KNX bus and Home Assistant (via IP Router).

- **PM2.5 Thresholds:** When PM2.5 exceeds **50–100 µg/m³**:
  - Automatically close all motorized skylights.
  - Stop all whole-house and attic exhaust fans.
  - Signal HVAC system to switch to 100% recirculation mode (via Modbus/Intesis gateway).
- **Weather Logic:**
  - **High Wind/Rain:** Trigger safety closing of skylights and adjust shading positions.
  - **Brightness/Global Radiation:** Orchestrate shading and ventilation for passive cooling and glare control.
  - **GPS Sun Position:** Real-time tracking for automated "Sun Following" blind logic.
- **System Bridging:** Real-time objects are pushed from KNX to the Linux NUC for high-level AI logic and SmartThings bridging.